

DEMO COURSE SYLLABUS

Coastal Geography

GEOS 340 – Section 001
Fall, 2025 · Mississippi State University

Faculty Information:		Teaching Assistant:
Instructor: Hafez Ahmad, PhD	Class Meetings: M/W/F 10:00 – 10:50 AM + Lab/Field: T 2:00 – 3:50 PM	Name: TA Name, PhD Student
Office Location: [Office Location]	Course Format: In-Person Lecture + In-Person Lab/Field Session (Zoom option available for excused absences)	TA Office: [TA Office]
Office Hours: M: 11:00 AM – 12:00 PM (In-person or Zoom) W: 10:30 – 11:30 AM (In-person or Zoom)	Room: [Office Room] Lab: [Lab Room]	TA Office Hours: TBA — Posted on Canvas
Email: ha463@cornell.edu Phone: 662-312-8601	Credits: 4 credit hours (3 lecture + 1 lab/field)	TA Email: ta.email@university.edu

Course Description

Coastal Geography examines the physical, environmental, social, and spatial processes that shape coastal environments and human communities. The course introduces students to coastal landforms, sediment transport, waves, tides, currents, sea-level change, estuaries, wetlands, barrier islands, deltas, coastal hazards, resource use, and human modification of coastal systems.

A central theme of the course is the interaction between coastal processes and human activities. Students will examine how urbanization, ports, tourism, fisheries, agriculture, engineering structures, wetland loss, and climate change influence coastal landscapes and communities.

The U.S. Southern Coast will serve as an integrated regional case-study area throughout the semester. Examples will include the Gulf Coast of Texas, Louisiana, Mississippi, Alabama, and Florida, as well as the Atlantic coast of Florida, Georgia, South Carolina, and North Carolina. Particular attention will be given to the Mississippi River Delta, Louisiana wetlands, Mississippi Sound, Mobile Bay, Apalachicola Bay, Florida Bay, the Florida Keys, Atlantic and Gulf barrier islands, and rapidly urbanizing coastal areas.

The course also introduces students to modern geospatial and Earth-observation approaches for understanding coastal environments, including GIS, satellite remote sensing, digital elevation models, aerial imagery, coastal-change mapping, and basic spatial analysis.

Course Prerequisites

There are no formal prerequisites. Previous coursework in physical geography, environmental science, geology, oceanography, or GIS may be helpful but is not required.

Credit Hours: 4 credit hours (3 hours lecture + 1 hour lab/field session)

Learning Objectives

By the end of this course, students will be able to:

- Explain the major physical processes that shape coastal landscapes.
- Identify and interpret major coastal landforms and environments.
- Explain the roles of waves, tides, currents, storms, sediment transport, and sea-level change in coastal evolution.
- Describe the ecological and geographic importance of estuaries, wetlands, mangroves, barrier islands, deltas, and coastal lagoons.
- Analyze interactions between human development and coastal processes.
- Explain major coastal hazards, including hurricanes, storm surge, coastal flooding, erosion, subsidence, and sea-level rise.
- Evaluate coastal management strategies and their environmental and socioeconomic consequences.
- Use maps, GIS, satellite imagery, and other spatial datasets to investigate coastal geographic problems.
- Interpret coastal change using historical imagery, elevation data, and environmental observations.
- Apply geographic concepts to real-world coastal problems in the U.S. Southern Coast.
- Compare coastal environments with different geographic and climatic settings.
- Communicate coastal geographic information using maps, figures, written analysis, and presentations.

Instructional Methodology | Course Format

This course follows a hybrid-friendly delivery model combining in-person lectures and laboratory/field sessions with supplementary online materials via Canvas. Synchronous class sessions are held at the times listed above; attendance is expected.

Technology Requirements

- Reliable high-speed internet connection (cable, DSL, or equivalent broadband)
- Modern web browser: Google Chrome (recommended), Firefox, or Safari latest version
- Adobe Acrobat Reader (free download at get.adobe.com/reader/)
- Microsoft Word, Excel, and PowerPoint (available free through your university's Office 365 license)
- Speakers or headphones for watching recorded lecture supplements and tutorial videos
- Webcam and microphone recommended for virtual office hours via Zoom

GIS Software

All lab assignments use ArcGIS Pro and QGIS with coastal datasets. Detailed installation instructions will be provided during Week 1 through Canvas.

- ArcGIS Pro is available through the [University] student license (Windows only). Students with Mac computers may use the GIS computer lab or set up Windows via Boot Camp/virtualization (instructions provided)
- QGIS 3.x LTR free, open-source, available for Windows, Mac, and Linux at qgis.org
- Anaconda Python software, open source, available for windows , and Mac
- Lab computers with ArcGIS Pro are available in [Building, Room] you will have ID badge access throughout the semester.

Data Storage

GIS work generates large files. Always save your projects to an external USB drive (5 GB or larger recommended) or a cloud service (OneDrive, Google Drive). Regularly back up all GIS project files the instructor is not responsible for work lost due to software crashes or hardware failure.

Canvas LMS

All course materials lecture slides, lab instructions, datasets, announcements, grades, and Zoom links are posted on Canvas. Log in using your university NetID and password.

Required Textbook & Software

Textbook

1. Davidson-Arnott, R., Bauer, B., & Houser, C. (2019). Introduction to Coastal Processes and Geomorphology, 2nd ed.
2. Davis, R. A., Jr., & FitzGerald, D. M. (2020). Beaches and Coasts (2nd ed.). Wiley.
3. Finkl, C. W. (Ed.). (2013). Coastal Hazards (Vol. 6). Springer

Additional readings, journal articles, and regional case-study materials will be posted on Canvas each week.

Note: A later edition is acceptable as a backup. Confirm with the instructor before purchasing an older edition.

Software (All Provided at No Cost to Students)

- ArcGIS Pro 3.x — [University] site license (Windows). Setup instructions: Canvas → Week 1 Module
- QGIS 3.x LTR — free download at qgis.org (Windows / Mac / Linux)
- All lab and case-study datasets provided through Canvas at no cost, including NOAA, USGS, and Sentinel/Landsat coastal datasets.

Performance-Based Assessments — Grading

Students achieve the course learning outcomes through assigned readings, weekly discussion participation, hands-on GIS lab and case-study exercises, three examinations, and a final research project.

Component	Points/Weight	% of Grade
Online Discussions / Coastal Case Study Reflections (4 forums)	20 pts total	5%
Coastal GIS Lab Assignments (9 labs)	180 pts total	40%
Exam 1 (Coastal Processes & Geomorphology)	45 pts	10%
Exam 2 (Estuaries, Wetlands & Deltas)	45 pts	10%
Exam 3 (Hazards, Human Modification & Ecosystems)	67 pts	15%
Final Coastal GIS Research Project	90 pts	20%
TOTAL	450 pts	100%

Online Discussions | Reading Reflections [5%]

There will be four discussion forums, each tied to a Gulf Coast or southeastern Atlantic case study, designed to ignite learning and peer engagement. For each forum, you will contribute two posts:

- An original response to the instructor's prompt (100–150 words), due Thursday (R) by 11:59 PM
- A thoughtful, substantive reply to at least two classmates (100–150 words each), due Sunday (Su) by 11:59 PM

Each contribution is assessed on factual accuracy, connection to module concepts, and depth of engagement. Grading rubric: Expected (5 pts) | Sufficient (4 pts) | Insufficient (3 pts) | Not Completed (2 pts).

GIS Lab Assignments [40%]

Nine GIS lab assignments are distributed throughout the semester, each designed to build practical geospatial skills applied to coastal systems along the U.S. Gulf and southeastern Atlantic coasts. Labs are conducted during the weekly lab/field session (Tuesday, 2:00–3:50 PM) and are due by the end of each lab week (Sunday, 11:59 PM). Skills developed include:

- Data Analysis and Visualization: transforming raw coastal geographic data (bathymetry, shoreline, wetland extent, imagery) into informative maps and charts
- Spatial Modelling: understanding relationships among coastal geographic features (shoreline change, marsh migration, storm surge) and predicting spatial patterns
- Problem Solving: applying GIS tools to answer real-world coastal management questions

Examinations [35%]

Three examinations cover the assigned textbook readings, lecture notes, and case-study materials. Exam formats include multiple-choice, definitions, fill-in-the-blank, and short-answer questions. Exams are administered online through Canvas within a designated time window.

- Exam 1 — 10% | Covers coastal processes, geomorphology, waves, tides, and barrier islands (Weeks 1–6)
- Exam 2 — 10% | Covers estuaries, coastal wetlands, and deltas (Weeks 7–9)
- Exam 3 — 15% | Covers coastal hazards, human modification, and coastal ecosystems (Weeks 10–13)

Final GIS Research Project [20%]

The end-of-semester project requires students to conduct an original GIS-based coastal research investigation, drawing on a Gulf Coast or southeastern Atlantic case study wherever possible. Project requirements:

- Incorporate GIS techniques; utilize GIS processing and analysis methods covered in class and lab assignments
- Address a research question; formulate a clear coastal geography question that can be answered using GIS and remote sensing methods learned in this course.
- Include a presentation and written report: present findings in class and submit a comprehensive written report detailing research design, methodology, results, and conclusions.

Full project guidelines will be distributed in Week 10. Students may complete the project individually or as a two-person team. Project topics must be approved by the instructor.

Grading Scale

Final grades will be assigned according to the following distribution:

Grade	Range	Grade	Range	Grade	Range
A	93–100	A–	90–92	B+	87–89
B	83–86	B–	80–82	C+	77–79
C	70–76	D	60–69	F	< 60

Note

There is no C– grade at Mississippi State University. A grade of 70% is the minimum for a C. Students earning below 60% will receive an F.

Policy on Missed Exams and Assignments

- All due dates are included in the course calendar on Canvas. Due times are based on Central Time (CT) — adjust accordingly if you are in a different time zone.
- Students have two "flex days" to submit up to two assignments up to 2 days late with no penalty. After the flex days are used, a 5-point-per-day late penalty applies. Assignments will not be accepted more than 5 days after the original deadline and will receive a zero.
- There are no make-ups for Discussion posts after the due date has passed.
- Make-up examinations are granted only in cases of documented personal illness, university-excused absences (e.g., varsity athletics, religious holidays), or other extraordinary circumstances at the instructor's discretion. If you know in advance that you cannot sit an exam, contact the instructor at least one week before the exam date.

Course Schedule / Calendar

Students should complete assigned readings before the first lecture of each week. Each week includes a dedicated Southern U.S. Coastal Focus column highlighting a regional case study. This schedule may be adjusted as necessary to accommodate weather, field conditions, current coastal events, or instructional needs.

Wk	Dates	Topic / Lecture	Southern U.S. Coastal Focus	Laboratory / Activity
1	Aug 18–24	Introduction to Coastal Geography	Overview of the Gulf and Atlantic coasts of the southern United States; major coastal environments	Coastal map orientation; identify major U.S. Southern Coast regions
2	Aug 25–31	Coastal Systems: Processes, Scales & Energy	Gulf of Mexico vs. Atlantic Ocean; regional differences in wave climate and coastline orientation	Coastal system conceptual model; Discussion 1 (R/Su)
3	Sep 1–7	Waves, Wave Energy & Longshore Transport	Texas and Florida beaches; wave-driven sediment transport	Calculate/interpret wave characteristics; shoreline sediment exercise
4	Sep 8–14	Tides, Currents & Coastal Circulation	Tidal regimes of Louisiana, Mississippi, Alabama, Georgia, and the Carolinas	Analyze tide-gauge data; tidal-range mapping
5	Sep 15–21	Coastal Sediment & Shoreline Change	Barrier-island sediment dynamics along the Gulf and Atlantic coasts	Historical shoreline-change analysis using aerial imagery; Discussion 2 (R/Su)
6	Sep 22–28	Beaches, Dunes & Barrier Islands	Padre Island, Dauphin Island, Mississippi barrier islands, Outer Banks	Barrier-island geomorphology mapping; Exam 1 (Weeks 1–6)
7	Sep 29 – Oct 5	Estuaries & Coastal Lagoons	Mississippi Sound, Mobile Bay, Galveston Bay, Tampa Bay, and Florida Bay	Estuary classification and salinity-gradient analysis
8	Oct 6–12	Coastal Wetlands, Marshes & Mangroves	Louisiana wetlands, Mississippi Sound marshes, Georgia salt marshes, and Florida mangroves	Wetland mapping using satellite imagery; Discussion 3 (R/Su)
9	Oct 13–19	Deltas & River–Coast Interactions	Mississippi River Delta as a major case study; sediment supply, subsidence, land loss, and delta evolution	Mississippi Delta change analysis using DEMs and satellite imagery; Exam 2 (Weeks 7–9)
10	Oct 20–26	Hurricanes, Storm Surge & Coastal Hazards	Katrina, Rita, Ike, Harvey, Michael, Ian, and other Southern Coast hurricanes	Storm-track and storm-surge mapping; Project Topics Due
11	Oct 27 – Nov 2	Sea-Level Rise, Subsidence & Coastal Flooding	Louisiana subsidence; Florida and Charleston vulnerability; recurrent tidal flooding	Analyze sea-level and elevation datasets; Discussion 4 (R/Su)
12	Nov 3–9	Human Modification of Coastal Environments	Ports, seawalls, levees, navigation channels, beach nourishment, and coastal development	Compare engineered vs. natural shorelines

13	Nov 10–16	Coastal Ecosystems & Environmental Change	Gulf Coast fisheries, oyster reefs, seagrass, coral reefs, mangroves, and salt marshes	Coastal habitat classification and change detection; Exam 3 (Weeks 10–13)
14	Nov 17–23	Coastal Urbanization, Tourism & Socioeconomic Geography	Miami, Tampa, New Orleans, Houston/Galveston, Charleston, and other rapidly developing coastal cities	Coastal population and development mapping
15	Nov 24–30	Coastal Management, Restoration & Adaptation	Louisiana Coastal Master Plan, Mississippi/Alabama restoration, Florida resilience planning, and Atlantic Coast management	Evaluate competing coastal-management strategies
16	Dec 1–7	Future Coasts & Student Presentations	Climate change, sea-level rise, adaptation, resilience, and the future of the Southern U.S. Coast	Final project presentations; Final Project Due
FINAL	TBD	Final Examination Period	—	—

Lab due dates

All labs are due by Sunday, 11:59 PM CT at the end of the week they are assigned. Discussion initial posts (R) are due Thursday, 11:59 PM; replies (Su) are due Sunday, 11:59 PM.

University Policies & Student Support

Email Communication

Students are responsible for checking their official university email account regularly. All courses will be sent to your university email. The instructor will not respond to emails sent from personal (non-university) accounts. Please include "[Course Number]" in the subject line of all emails and begin with a proper greeting.

Academic Integrity

Mississippi State University operates under an Academic Integrity Policy. All work submitted must be your own. Violations, including cheating, plagiarism, and unauthorized collaboration, will be reported to the appropriate academic integrity office and may result in a failing grade for the course. See: [honor code website]

Disability Accommodations

Mississippi State University is committed to providing equitable access to learning for all students. Students with documented disabilities requiring academic accommodation should register with Student Disability Services and provide the instructor with an accommodation letter within the first two weeks of class.

Student Disability Services · [Location] · Phone: [Phone] · Email: [email] · Website: [website]

Diversity, Equity & Inclusion

Mississippi State University is committed to providing an educational environment free from discrimination on the basis of race, gender, sexual orientation, disability, religion, national origin, age, or any other protected characteristic. All students are expected to engage respectfully with diverse perspectives. Discriminatory or harassing behavior will not be tolerated.

Responsible Use of Technology

Students must follow Mississippi State University's Acceptable Use Policy for computing and network resources. This includes ethical and legal use of software, data, and online communications.

Title IX / Sexual Misconduct

As a faculty member, the instructor is a Responsible Employee and is required to report disclosures of sexual assault, interpersonal violence, or stalking to the university's Title IX Coordinator. If you wish to speak confidentially, contact the Student Counseling Center or University Counseling and Psychological Services (CAPS).

FERPA / Student Privacy

Mississippi State University complies fully with the Family Educational Rights and Privacy Act (FERPA). Student educational records are protected and will not be disclosed without the student's written consent, except as allowed by law.

Absences & Accommodations

- Religious Observances Contact the instructor at least one week in advance if a religious observance will conflict with a class session or deadline. Accommodation will be provided.
- Illness Students who miss multiple sessions due to prolonged illness should seek medical care and provide documentation to the Dean's Office, which will coordinate with the instructor. Extended absences may necessitate medical withdrawal.
- All students, regardless of the reason for absence, remain responsible for all assigned material and must arrange to make up missed work when accommodation is approved.

Student Responsibilities

- Stay engaged with readings, lectures, lab exercises, and discussion deadlines.
- Respect classmates and the instructor in all course interactions in person and online
- Communicate professionally in all written and verbal interactions
- Use your official university email for all course communications
- Back up GIS project files regularly software crashes are unpredictable

Student Support Services

- Counseling and Psychological Services (CAPS): [caps.university.edu]
- Student Disability Services: [sds.university.edu]
- The Writing Center: [writing.university.edu]
- University Career Center: [career.university.edu]
- Library & Research Services: [libraries.university.edu]
- IT Help Desk: [it@university.edu] · Phone: [Phone]

A Note from the Instructor

Coastal geography is a hands-on, visual, and deeply consequential field — especially along a coastline as dynamic as the U.S. Gulf and southeastern Atlantic coast. Whether you are a geography major or come from an entirely different field, I am here to make sure you leave this course feeling capable and confident working with coastal data, from a beach profile to a satellite image. Ask questions early, back up your work often, and do not hesitate to reach out during office hours. I look forward to a great semester.

Syllabus current as of August 2026 · Subject to revision with advance notice to students.